DRIVING EMBEDDED EXCELLENCE

RTA-SUM - State of Health Manager (SOH) 8.0.1_Pr6

User Guide

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Contents

1 Introduction **4**
1.1 Definitions and Abbreviations . . . 4
1.2 Module Overview . . . 5
1.3 References . . . 6
1.4 User Guide Revision History . . . 6

2 Integration Notes **8**
2.1 Integration into a Project . . . 8
2.2 Module Dependencies . . . 8
2.3 Configuring SOH . . . 8
2.4 Generating Artifacts . . . 8
2.5 Configuring the RTE . . . 8
2.6 Configuring OS . . . 9
2.6.1 OsCounter . . . 10

3 Module configuration **12**
3.1 Configuration Parameters . . . 12
3.1.1 SOHGeneral . . . 12
3.1.2 SOHConfigSet . . . 14
3.2 Configuration Hints . . . 21
3.2.1 SOH Initialization . . . 21
3.2.2 Memory Types . . . 21
3.2.3 Main FSM . . . 23
3.2.4 Time Manager . . . 23
3.2.5 Time Window Filter . . . 24
3.2.6 Reliability Monitor . . . 24
3.2.7 DID Counters . . . 34
3.2.8 WdgM . . . 36
3.2.9 RID 0x02C0 Clear SOH Logs . . . 36
3.2.10 RTE RteSwComponentType configuration . . . 37

4 Generic Component Interface **38**

5 Contact, Support and Problem Reporting **43**
5.1 Technical Support . . . 43
5.2 ETAS Global Headquarters . . . 43
5.3 ETAS Local Sales and Support Offices . . . 43

1 Introduction

This document is intended to describe the main steps required for the integration and configuration of RTA-SUM SOH module.

1.1 Definitions and Abbreviations

Term/Abbreviation	Definition
API	Application Program Interface
BSW	Basic Software
BswM	Basic Software Mode Manager
CAN	Controller Area Network
Dcm	Diagnostic Communication Module
DID	Data IDentifier
DTC	Diagnostic Trouble Code
DIAG	Diagnostic Application Services
ECU	Electronic Control Unit
EcuM	ECU State Manager
ERRH	Error Handling Module
LIN	Local Interconnect Network
LNM	LIN Network Manager
NVRAM	Non Volatile Random Access Memory
NvM	NVRAM Manager
OAW	Open Architecture Ware
OS	Operating System (OSEK compliant)
PMS	Power Mode Slave
PNC	Partial Network Coordinator
RTE	Run Time Environment
RTOS	Real Time OS
SOH	State Of Health Manager
SSM	Signal Status Monitor
SSC	Security Service Coordinator
SUM	Standard Utility Module
SUSD	Startup Shut down Coordinator
SW	Software
SW-C	Software Component

Term/Abbreviation	Definition
TCS	Timing Client Service
VIP	Vehicle Intelligence Platform
WdgM	Watchdog Manager

1.2 Module Overview

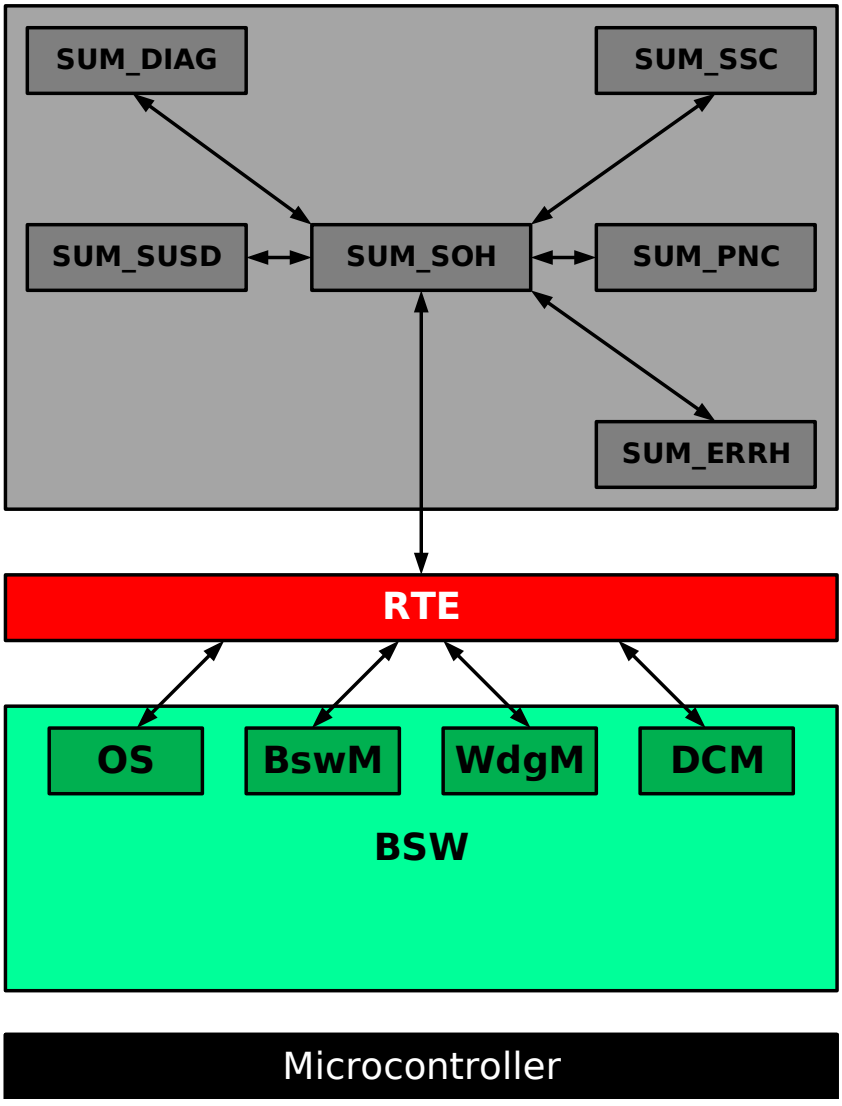


Figure 1.1: Layers

The SUM_SOH is a software module in the application layer of the standardized architecture of AUTOSAR.

SUM_SOH implements the following functionalities:

- Reset reliability monitor.
- Memory Status reliability monitor.
- Platform reliability monitor.
- Communication reliability monitor.
- Watchdog Manager reliability monitor.
- Provide monitored events data via DIDs 0xF08F to 0xF093 (DID 0xF094 is implemented by ERRH).
- Clear monitored events logs via RID.

1.3 References

Document Name	Description	Version	Released
SDV-8002	AUTOSAR Standard Utility Modules Specification	27.27.180.3p0	04/04/2025
SDV-6001	Diagnostic Infrastructure Specification	27.27.180.3p0	04/09/2025
CYS5402	SecOC and SOME IP Deployment Requirements	27.27.180.3p0	03/04/2025

1.4 User Guide Revision History

Version	Description of Changes	Author
8.0.0_Proto1	First prototype release of RTA-SUM SOH for SDV 2.0 Migration to AUTOSAR 20.11 (RTA-CAR 12.0).	SUM Development Team
8.0.0_Proto2	Second prototype release of RTA-SUM SOH for SDV 2.0	SUM Development Team
8.0.0_Proto3	Third prototype release of RTA-SUM SOH for SDV 2.0	SUM Development Team
8.0.0_Proto4	Fourth prototype release of RTA-SUM SOH for SDV 2.0	SUM Development Team
8.0.0_Proto5	Fifth prototype release of RTA-SUM SOH for SDV 2.0	SUM Development Team
8.0.0_Proto5_1	Incremental prototype release of RTA-SUM SOH for SDV 2.0	SUM Development Team
8.0.0_Proto6	Sixth prototype release of RTA-SUM SOH for SDV 2.0	SUM Development Team

Version	Description of Changes	Author
8.0.1_Pr1	First prototype release of RTA-SUM SOH 8.0.1 for SDV 2.0 Document version updated.	SUM Development Team
8.0.1_Pr2	Second prototype release of RTA-SUM SOH 8.0.1 for SDV 2.0 Added Section 2.6.1	SUM Development Team
8.0.1_Pr3	Third prototype release of RTA-SUM SOH 8.0.1 for SDV 2.0 Document version updated.	SUM Development Team
8.0.1_Pr4	Fourth prototype release of RTA-SUM SOH 8.0.1 for SDV 2.0 Document version updated.	SUM Development Team
8.0.1_Pr5	Fifth prototype release of RTA-SUM SOH 8.0.1 for SDV 2.0 Updated section 3.1.1 and 3.2.1 Updated section 4 to align GetMec port interface.	SUM Development Team
8.0.1_Pr6	Sixth prototype release of RTA-SUM SOH 8.0.1 for SDV 2.0 Updated section 3.2.2.	SUM Development Team

2 Integration Notes

2.1 Integration into a Project

There is no need for standard installation procedure for SOH. The folder itself contains all the necessary artifacts to integrate into a project. Dropping the complete folder structure into a BSW configuration project will enable the user to configure SOH and generate all the necessary artifacts.

Refer to RTA-SUM Knowledge Base

2.2 Module Dependencies

When connecting the Dcm to the port interface `DataServices<SOH>Data<n>` the DID counters `DID 0xF08F, DID 0xF090, DID 0xF091, DID 0xF092, DID 0xF093` will be updated through operation invoked event Runnables.

If OS Status Monitoring is enabled (look 3.1.1), please Make reference to 2.6.

2.3 Configuring SOH

Once the complete package has been added to a BSW project, an additional module SOH should appear. From here you can open and begin configuring the module. Specific description of each configurable parameter is available in the “Description” view of ISOLAR-A with BCT add-on when a parameter is selected.

2.4 Generating Artifacts

Generating the C/H/SWCD files from your configuration is similar to other BSW modules delivered from ETAS. The Code Generation perspective of the ISOLAR-A with BCT add-on, will present the user with a dialog to select specific module within the project to generate.Either a Partial Build or Full Build can be used, depending on the needs of the user.In case of any configuration errors present in SOH, the console window will output a description of any errors found during generation. Further, when the user is configuring the module, the Problems Log will show any active errors in the configuration of SOH. A third visual aid is the appearance of red X’s in the configuration view over items which are invalid. Simply hovering over these X’s will provide the user a description of the problem. These are static validations.

2.5 Configuring the RTE

After the user has configured and generated the artifacts for SOH, the output SWCD file (SUM_SOH_Swcd.arxml) can be used for the RTE configuration. This is a complete SWC description per the AUTOSAR standard of all the users configured ports and interfaces required for the project specific configuration. This SWCD is a standard ASW component and not a module description. Therefore, there are no special considerations to be taken into account when configuring the RTE. It is recommended that if specific changes are required to the SWCD, the user not manually edit this file, rather update and regenerate the configuration with the required changes.

The minimum **RTA-RTE version** supported is the *7.3.0*.

Note: RTE command used for RTE generation

```
-err=xml -nts --samples=memmap --os-define-osenv=RTAOS40 --exclusive-area-
    optimization=disable --os-output-param=all --strict-unconnected-rport-check=
    off
```

2.6 Configuring OS

The following integration notes about the OS configuration consider RTA-OS version 6.0.0 by ETAS for reporting examples, templates and Samples but it can be adopted for each RTOS.
SUM_SOH module shall monitor the OS Errors occurrences. This implicates that SUM_SOH has got a way to be notified by OS about the OS Error Occurrence. The notification between OS and SUM_SOH is possible thanks the following Callbacks functions:

- `ErrorHook`
- `ProtectionHook`
- `SOH_OS_Protection_ErrorHook_CallBack`: look 3.2.6

The `ErrorHook` and `ProtectionHook` Callbacks are customized functions and **are in the responsibility of the Integrator/User, according to its OS specification**;

ErrorHook/ProtectionHook Callbacks shall be called by OS each time that one of its API produces an Error, in order to identify the type of OS Error and handle it for ensuring the right System Running.

To notify the SUM_SOH about the OS Error Occurrence it is necessary to call function `SOH_OS_Protection_ErrorHook_CallBack` within the ErrorHook and ProtectionHook, passing the ErrorType as function parameter to them. Figure 2.1 shows the intended sequences to report an OS Error to SUM_SOH.

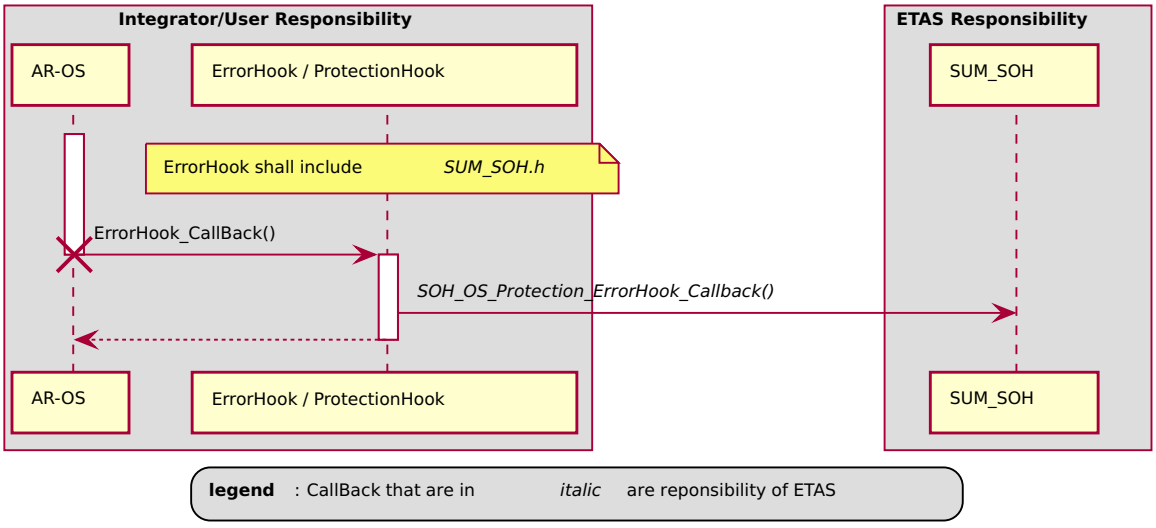


Figure 2.1: Callback sequence to report OS Error to SUM_SOH

2.6.1 OsCounter

In order to let the Main and Time Manager works properly, the OS Counter shall be configured as follows:

- `OsCounterMaxAllowedValue`: The max value that integrator can set is 4294967295 since the tick in the SOH are stored in a uint32;
- `OsSecondsPerTick`: Choose a value that can provide a good granularity w.r.t. the periodicity of the Runnable Runnable_SOH_Timing (configured in SOHGeneral/SOH_RunnnableSOHTimingPeriod);

An example of configuration is provided below:

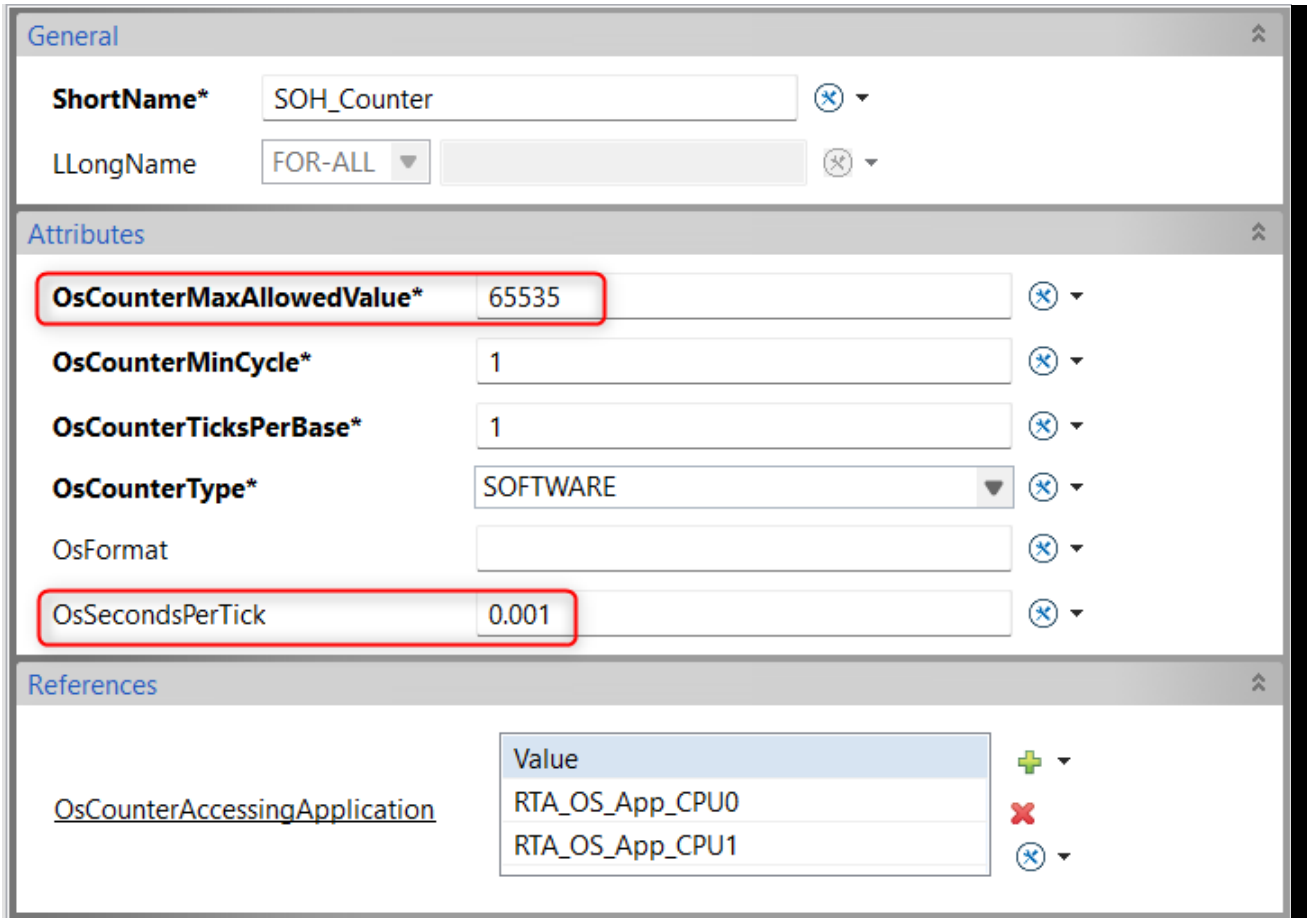


Figure 2.2: Example of OsCounter configuration needed.

The OsCounter configured SHALL be referenced in the SOHGeneral/Sum_SOH_Os_Counter_Ref as follow:

Please, take into account that the integrator can reference the timer referenced in the SSFM module, parameter Sum_Ssfm_Os_Counter_1ms_ref (in Sum_Ssfm_General container.)

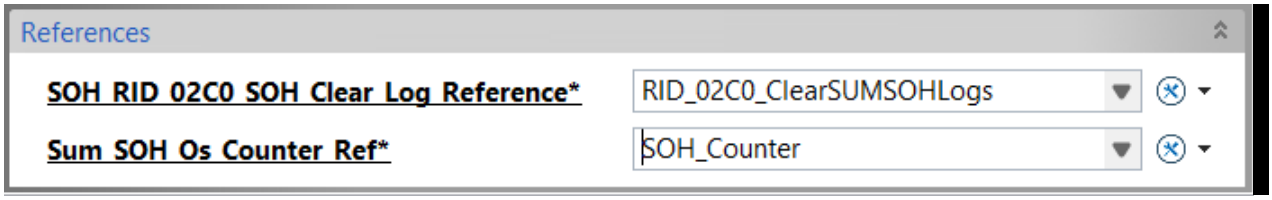


Figure 2.3: Reference to the OsCounter configured.

3 Module configuration

This section is intended to provide guidelines for RTA-SUM SOH configuration.

3.1 Configuration Parameters

3.1.1 SOHGeneral

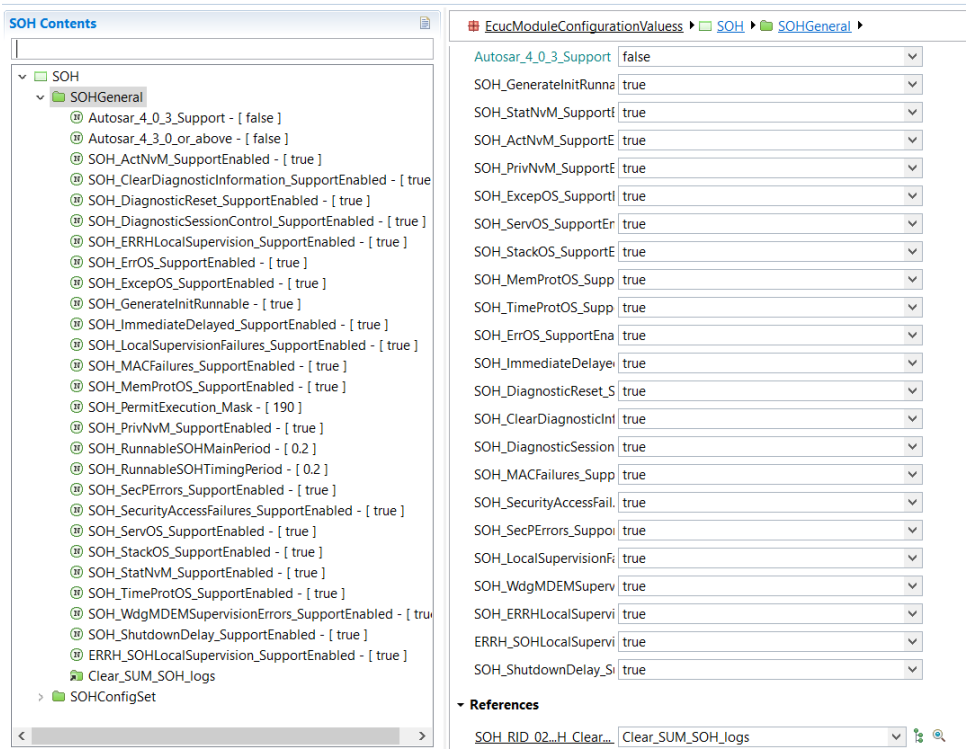


Figure 3.1: SOHGeneral container

SOH_PermitExecution_Mask
Mask that can be OR'd with the value supplied by PermitExecution to set any "don't care" bits to '1'.

SOH_RunnableSOHMainPeriod
This parameter specifies the period (in seconds) of the Runnable_SOH_Main, the one that implements the FSM of the module.

SohGenerateInitRunnable
This parameter is used to generate a periodic runnable entity to call SOH_init function only once.

SohInitFuncPeriodicity
This parameter specifies the time in milisecond for the periodicity of the runnable Runnable_SOH_Init.

SOH_StatNvM_SupportEnabled
This parameter is used to enable the Nvm Status Failure/Integrity Error class and the relative detector.

SOH_ActNvM_SupportEnabled
This parameter is used to enable the Nvm Action Failure Error class and the relative detector.

SOH_ExcepOS_SupportEnabled
This parameter is used to enable the OS Hardware Protection Error class and the relative detector.

SOH_ServOS_SupportEnabled
This parameter is used to enable the OS Service Protection / Trusted Function Error class and the relative detector.

SOH_StackOS_SupportEnabled
This parameter is used to enable the OS Stack Fault class and the relative detector.

SOH_MemProtOS_SupportEnabled
This parameter is used to enable the OS Memory Protection Failure Error class and the relative detector.

SOH_TimeProtOS_SupportEnabled
This parameter is used to enable the OS Time Protection Error class and the relative detector.

SOH_ErrOS_SupportEnabled
This parameter is used to enable the OS Other OS Errors Error class and the relative detector.

SOH_PrivNvM_SupportEnabled
This parameter is used to enable the Nvm Privilege class and the relative detector.

SOH_RID_02C0_SOH_Clear_Log_Reference Reference to DCM RID 0x02C0 SOH Clear Logs

SOH_ImmediateDelayed_SupportEnabled
It allows to enable (disable) for Immediate Reset Detector and Delayed Reset Detector in Reset Reliability monitor.

SOH_DiagnosticReset_SupportEnabled
It allows to enable (disable) for Diagnostic Reset Detector in Reset Reliability monitor.

SOH_ClearDiagnosticInformation_SupportEnabled
It allows to enable (disable) for Clear Diagnostic Information Accepted Request Detector and Clear Diagnostic Information Rejected Request Detector in Communication Reliability Monitor.

SOH_DiagnosticSessionControl_SupportEnabled
It allows to enable (disable) for Diagnostic Session Control Request Detector and Accepted Diagnostic Session Control Request Detector in Communication Reliability Monitor.

SOH_SecurityAccessFailures_SupportEnabled
It allows to enable (disable) for SecurityAccess (Service 0x27) Failures Detector in Communication Reliability Monitor.

SOH_MACFailures_SupportEnabled
It allows to enable (disable) for MAC Failures Types 1 and 2 Detector and MAC Failures Type 3 Detector in Communication Reliability Monitor.

SOH_MACFailures_SupportEnabled
It allows to enable (disable) for SecP Errors Detector in Communication Reliability Monitor.

SOH_LocalSupervisionFailures_SupportEnabled
It allows to enable (disable) for Local Supervision Failures Detector in Watchdog Manager Reliability Monitor.

SOH_WdgMDEMSupervisionErrors_SupportEnabled
It allows to enable (disable) for WdgM DEM Supervision Errors Detector in Watchdog Manager Reliability Monitor.

SOH_ERRHLocalSupervision_SupportEnabled
It allows to enable (disable) for SUM_ERRH Local Supervision Detector in Watchdog Manager Reliability Monitor.

ERRH_SOHLocalSupervision_SupportEnabled
It allows to enable (disable) for SUM_SOH Local Supervision Detector implemented by SUM_ERRH.

SOH_ShutdownDelay_SupportEnabled
It allows to enable (disable) for Shutdown Delay Detector in Platform Reliability Monitor.

SOH_RunnableSOHTimingPeriod
This parameter specifies the period (in seconds) of the runnable Runnable_SOH_Timing which implements the time manager for the Time Window Filter and the Shutdown Delay Detector.

3.1.2 SOHConfigSet

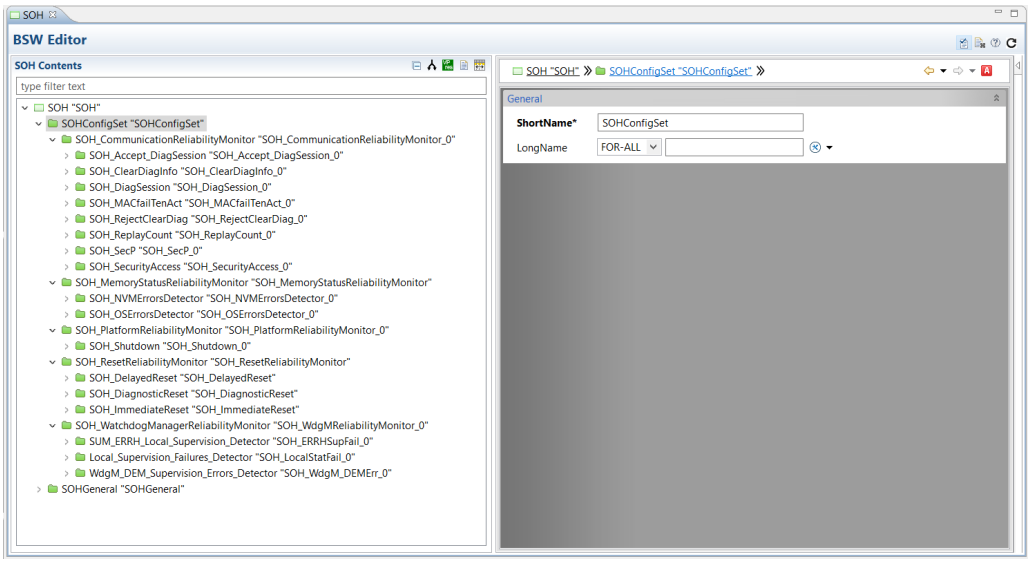


Figure 3.2: SOHConfigSet container

SOH_ResetReliabilityMonitor

SOH_ImmediateReset
SUM_SOH_ImmeReset_Thresh_Init
This parameter specifies the initial value of SUM_SOH_ImmeReset_Thresh

SUM_SOH_ImmeReset_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_ImmeReset_TW_Thresh

SOH_NvMImmResetDID_Reference
Reference to the NVM DID Block

SOH_DID_F08F_ImmResetData_Reference
Reference to DCM ImmReset Data

SOH_DelayedReset
SUM_SOH_DelayedReset_Thresh_Init
This parameter specifies the initial value of SUM_SOH_DelayedReset_Thresh

SUM_SOH_DelayedReset_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_DelayedReset_TW_Thresh

SOH_NvMDelayedResetDID_Reference
Reference to the NVM DID Block

SOH_DID_F08F_DelayedResetData_Reference
Reference to DCM DelayedReset Data

SOH_DiagnosticReset
SUM_SOH_DiagReset_Thresh_Init
This parameter specifies the initial value SUM_SOH_DiagReset_Thresh

SOH_NvMDiagResetCntr_Reference
Reference to the NVM Counter Block

SOH_NvMDiagResetDID_Reference
Reference to the NVM DID Block

SOH_DID_F08F_DiagResetData_Reference
Reference to DCM DiagReset Data

SOH_CommunicationReliabilityMonitor

SOH_ClearDiagInfo
SUM_SOH_ClearDiagInfo_Thresh_Init
This parameter specifies the initial value of SUM_SOH_ClearDiagInfo_Thresh

SOH_NvMClearDiagInfoDID_Reference
Reference to the NVM DID Block

SOH_DID_F092_ClearDiagInfoDID_Reference
Reference to DCM Clear Diag Infomation Data

SOH_Reject_ClearDiag
SUM_SOH_Reject_ClearDiag_Thresh_Init
This parameter specifies the initial value of SUM_SOH_Reject_ClearDiag_Thresh

SOH_NvMRejectClearDiagDID_Reference
Reference to the NVM DID Block

SOH_DID_F092_RejectClearDiagDID_Reference
Reference to DCM Reject Clear Diag Data

SOH_Accept_DiagSession

SUM_SOH_Accept_DiagSession_Thresh_Init
This parameter specifies the initial value of SUM_SOH_Accept_DiagSession_Thresh

SOH_NvMAccept_DiagSessionDID_Reference
Reference to the NVM DID Block

SOH_NvMAccept_DiagSessionCntr_Reference
Reference to the NVM DID Block

SOH_DID_F092_Accept_DiagSessionDID_Reference
Reference to DCM Accept Diag Session Data

SOH_DiagSession
SUM_SOH_DiagSession_Thresh_Init
This parameter specifies the initial value of SUM_SOH_DiagSession_Thresh

SOH_NvMDiagSessionDID_Reference
Reference to the NVM DID Block

SOH_NvMDiagSessionCntr_Reference
Reference to the NVM DID Block

SOH_DID_F092_DiagSessionDID_Reference
Reference to DCM Diag Session Data

SOH_SecurityAccess
SUM_SOH_SecurityAccess_Thresh_Init
This parameter specifies the initial value of SUM_SOH_SecurityAccess_Thresh

SOH_NvMSecurityAccessDID_Reference
Reference to the NVM DID Block

SOH_NvMSecurityAccessCntr_Reference
Reference to the NVM DID Block

SOH_DID_F092_SecurityAccessDID_Reference
Reference to DCM Security Access Data

SOH_MACfailTenAct
SUM_SOH_MACfail_TenAct_Thresh_Init
This parameter specifies the initial value of SUM_SOH_MACfail_TenAct_Thresh

SUM_SOH_MACfail_TenAct_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_MACfail_TenAct_TW_Thresh

SOH_NvMMACfail_TenActDID_Reference
Reference to the NVM DID Block

SOH_DID_F092_MACfail_TenActDID_Reference
Reference to DCM MAC fail type 1 and 2 Data

SOH_ReplayCount
SUM_SOH_ReplayCount_Thresh_Init
This parameter specifies the initial value of SUM_SOH_ReplayCount_Thresh

SUM_SOH_ReplayCount_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_ReplayCount_TW_Thresh

SUM_SOH_ReplayCount_Start_Thresh_Init
This parameter specifies the initial value of SUM_SOH_ReplayCount_Start_Thresh

SOH_NvMReplayCountDID_Reference
Reference to the NVM DID Block

SOH_DID_F092_ReplayCountDID_Reference
Reference to DCM MAC fail type 3 Data

SOH_SecP
SUM_SOH_SecP_Thresh_Init
This parameter specifies the initial value of SUM_SOH_SecP_Thresh

SUM_SOH_SecP_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_SecP_TW_Thresh

SOH_NvMSecPDID_Reference
Reference to the NVM DID Block

SOH_DID_F092_SecPDID_Reference
Reference to DCM SecP Data

SOH_MemoryStatusReliabilityMonitor

SOH_NVMErrorsDetector

SOH_StatusNVM
SUM_SOH_StatNVM_Thresh_Init
This parameter specifies the initial value of SUM_SOH_StatNVM_Thresh

SUM_SOH_StatNVM_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_StatNVM_TW_Thresh

SOH_NvMActNVMDID_Reference
Reference to the NVM DID Block

SOH_DID_F090_StatNVM_Reference
Reference to DCM StatNVM Data

SOH_ActNVM
SUM_SOH_ActNVM_Thresh_Init
This parameter specifies the initial value of SUM_SOH_ActNVM_Thresh

SUM_SOH_ActNVM_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_ActNVM_TW_Thresh

SOH_NvMActNVMDID_Reference
Reference to the NVM DID Block

SOH_DID_F090_ActNVM_Reference
Reference to DCM ActNVM Data

SOH_PrivNVM

SUM_SOH_PrivNVM_Thresh_Init
This parameter specifies the initial value of SUM_SOH_PrivNVM_Thresh

SUM_SOH_PrivNVM_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_PrivNVM_TW_Thresh

SOH_NvMPrivNVMDID_Reference
Reference to the NVM DID Block

SOH_DID_F090_PrivNVM_Reference
Reference to DCM PrivNVM Data

SOH_OSErrorsDetector

SOH_ExcepOS

SUM_SOH_ExcepOS_Thresh_Init
Initial value of SUM_SOH_ExcepOS_Thresh

SUM_SOH_ExcepOS_TW_Thresh_Init
Initial value of SUM_SOH_ExcepOS_TW_Thresh

SOH_NvMExcepOSDID_Reference
Reference to the NVM DID Block

SOH_DID_F090_ExcepOS_Reference
Reference to DCM ExcepOS Data

SOH_ServOS

SUM_SOH_ServOS_Thresh_Init
Initial value of SUM_SOH_ServOS_Thresh

SUM_SOH_ServOS_TW_Thresh_Init
Initial value of SUM_SOH_ServOS_TW_Thresh

SOH_NvMServOSDID_Reference
Reference to the NVM DID Block

SOH_DID_F090_ServOS_Reference
Reference to DCM ServOS Data

SOH_StackOS

SUM_SOH_StackOS_Thresh_Init
Initial value of SUM_SOH_StackOS_Thresh

SUM_SOH_StackOS_TW_Thresh_Init
Initial value of SUM_SOH_StackOS_TW_Thresh

SOH_NvMServOSDID_Reference

Reference to the NVM DID Block

SOH_DID_F090_StackOS_Reference
Reference to DCM StackOS Data

SOH_MemProtOS

SUM_SOH_MemProtOS_Thresh_Init
Initial value of SUM_SOH_MemProtOS_Thresh

SUM_SOH_MemProtOS_TW_Thresh_Init
Initial value of SUM_SOH_MemProtOS_TW_Thresh

SOH_NvMMemProtOSDID_Reference
Reference to the NVM DID Block

SOH_DID_F090_MemProtOS_Reference
Reference to DCM MemProtOS Data

SOH_TimeProtOS

SUM_SOH_TimeProtOS_Thresh_Init
Initial value of SUM_SOH_TimeProtOS_Thresh

SUM_SOH_TimeProtOS_TW_Thresh_Init
Initial value of SUM_SOH_TimeProtOS_TW_Thresh

SOH_NvMTimeProtOSDID_Reference
Reference to the NVM DID Block

SOH_DID_F090_TimeProtOS_Reference
Reference to DCM TimeProtOS Data

SOH_ErrOS

SUM_SOH_ErrOS_Thresh_Init
Initial value of SUM_SOH_ErrOS_Thresh

SUM_SOH_ErrOS_TW_Thresh_Init
Initial value of SUM_SOH_ErrOS_TW_Thresh

SOH_NvMErrOSDID_Reference
Reference to the NVM DID Block

SOH_DID_F090_ErrOS_Reference
Reference to DCM ErrOS Data

SOH_WatchdogManagerReliabilityMonitor

Local_Supervision_Failures_Detector
SUM_SOH_LocalStatFail_Thresh_Init
This parameter specifies the initial value of SUM_SOH_LocalStatFail_Thresh

SUM_SOH_LocalStatFail_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_LocalStatFail_TW_Thresh

Supervised_Entity_Reference
This is a reference to a Supervised Entity container in the WdgM. The multiplicity is infinite. All the supervised entities (except the ones related to ERRH and SOH) should be referenced by using this parameter.

SOH_LocalSupervisionFailures_DID_counter_Reference
Reference to the NVM Block related to SUM_SOH_DID_LocalStatFail_Cntr

SOH_DID_F093_LocalStatFail_Reference
Reference to DCM LocalStatFail Data

WdgM_DEM_Supervision_Errors_Detector
SUM_SOH_WdgM_DEMErr_Thresh_Init
This parameter specifies the initial value of SUM_SOH_WdgM_DEMErr_Thresh

SUM_SOH_WdgM_DEMErr_TW_Thresh_Init
This parameter specifies the initial value of SUM_SOH_WdgM_DEMErr_TW_Thresh

SOH_WdgMDEMErr_DID_counter_Reference
Reference to the NVM Block related to SUM_SOH_DID_WdgM_DEMErr_Cntr

SOH_DID_F093_WdgMDEMErr_Reference
Reference to DCM WdgMDEMErr Data

SUM_ERRH_Local_Supervision_Detector
SOH_ERRHSupFail_Thresh_Init
This parameter specifies the initial value of SOH_ERRHSupFail_Thresh

SOH_ERRHSupFail_TW_Thresh_Init
This parameter specifies the initial value of SOH_ERRHSupFail_TW_Thresh

SOH_ERRHLocalSupervisionFailures_DID_counter_Reference
Reference to the NVM Block related to SUM_SOH_DID_ERRHSupFail_Cntr

SOH_DID_F093_ERRHSupFail_Reference
Reference to DCM ERRHSupFail Data

SOH_DID_F093_ERRH_Local_Supervision_Status_Reference
Reference to DCM ERRH Local Supervision Status Data

SOH_PlatformReliabilityMonitor

SOH_Shutdown
SUM_SOH_Shutdown_Thresh_Init
This parameter specifies the initial value of SUM_SOH_Shutdown_Thresh

SUM_SOH_ShutDown_Timer_Thresh_Init
This parameter specifies the initial value of SUM_SOH_ShutDown_Timer_Thresh

SOH_NvMShutdownDID_Reference
Reference to the NVM DID Block

SOH_NvMShutdownCntr_Reference
Reference to the NVM Counter Block

SOH_DID_F091_ShutdownDID_Reference
Reference to DCM Shutdown DID counter Data

3.2 Configuration Hints

3.2.1 SOH Initialization

To correctly initialize the module, the user can follow two possible paths depending on the configured value for the boolean parameter `SohGenerateInitRunnable` :

- If it is false, the user shall invoke the SUM_SOH initialization function `SOH_Init()`, which is an external function declared in the header file `SUM_SOH.h`. The BswM should invoke this function at initialization time through a BswMAction.
- If it is true, the user shall configure the `SohInitFuncPeriodicity` parameter, which specifies the periodicity of the `Runnable_SOH_Init` in ms, and schedule the init runnable in a lower periodicity OS task. In this case, a periodic runnable is generated to execute the internal `SOH_Init()` function.

3.2.2 Memory Types

SUM_SOH requires four different types of memory :

- RAM
- Non Cleared RAM
- Adaptable NVM
- Secure Cumulative NVM

The ECU shall have a NvM Module fulfilling the requirements for Adaptable NVM and Secure Cumulative NVM definitions.

One Block for each NvM metric shall be created (30 different blocks), each block shall fulfill the respective requirements for Adaptable NVM or Secure Cumulative NVM definitions.

Each NvM block shall have the **NvMRamBlockDataAddress** set to the relative RTE generated ArPim variable like in the figure 3.3

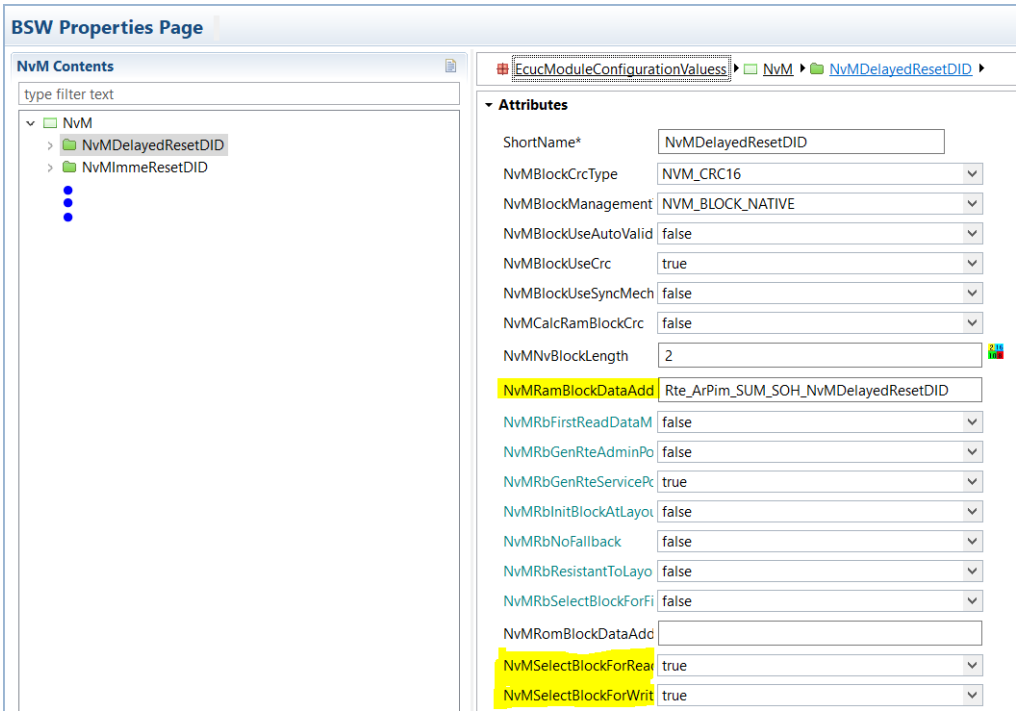


Figure 3.3: example of NvM block

For RAM and Non Cleared RAM types, the user shall use the MemMapGen generator to generate the `SUM_SOH_MemMap.h` file. This file will contain the necessary memory sections, defined in the SWCD (Figure: 3.4), as below:

- `SUM_SOH_SEC_CODE` used for Code.
- `SUM_SOH_SEC_VAR_POWER_ON_CLEARED_BOOLEAN` used for Non Cleared RAM boolean.
- `SUM_SOH_SEC_VAR_POWER_ON_CLEARED_16` used for Non Cleared RAM 16Bit.
- `SUM_SOH_SEC_VAR_POWER_ON_CLEARED_32` used for Non Cleared RAM 32Bit.
- `SUM_SOH_SEC_VAR_CLEARED_BOOLEAN` used for all the others boolean.
- `SUM_SOH_SEC_VAR_CLEARED_16` used for all the others 16Bit.
- `SUM_SOH_SEC_VAR_CLEARED_32` used for all the others 32Bit.
- `SUM_SOH_SEC_VAR_CLEARED_UNSPECIFIED` used for structures and array.

```
<RESOURCE-CONSUMPTION>
 <SHORT-NAME>Soh_ResourceConsumption_0</SHORT-NAME>
 <MEMORY-SECTIONS>
   <MEMORY-SECTION>
     <SHORT-NAME>VAR_CLEARED_BOOLEAN</SHORT-NAME>
   </MEMORY-SECTION>
   <MEMORY-SECTION>
     <SHORT-NAME>VAR_CLEARED_16</SHORT-NAME>
   </MEMORY-SECTION>
   <MEMORY-SECTION>
     <SHORT-NAME>VAR_CLEARED_32</SHORT-NAME>
   </MEMORY-SECTION>
   <MEMORY-SECTION>
     <SHORT-NAME>VAR_POWER_ON_CLEARED_BOOLEAN</SHORT-NAME>
   </MEMORY-SECTION>
   <MEMORY-SECTION>
     <SHORT-NAME>VAR_POWER_ON_CLEARED_16</SHORT-NAME>
   </MEMORY-SECTION>
   <MEMORY-SECTION>
     <SHORT-NAME>VAR_POWER_ON_CLEARED_32</SHORT-NAME>
   </MEMORY-SECTION>
   <MEMORY-SECTION>
     <SHORT-NAME>VAR_CLEARED_UNSPECIFIED</SHORT-NAME>
   </MEMORY-SECTION>
   <MEMORY-SECTION>
     <SHORT-NAME>CODE</SHORT-NAME>
   </MEMORY-SECTION>
 </MEMORY-SECTIONS>
</RESOURCE-CONSUMPTION>
<BEHAVIOR-REF DEST="SWC-INTERNAL-BEHAVIOR">/SUM_SOH/Component/SUM_SOH/SUM_SOH_IB</BEHAVIOR-REF>
```

Figure 3.4: Memory sections defined in the SWCD arxml file.

3.2.3 Main FSM

The main SOH Finite State Machine is implemented by the Runnable_SOH_Main. It manages three states:

- SOH_INIT_MODE
- SOH_NORMAL_MODE
- SOH_SHUTDOWN_MODE

To properly work it needs the following port connections :

- `SUSD_SOH_PermitExecution` with the SUSD
- `SOH_SUSD_ShutdownPermission` with the SUSD
- `DIAG_SOH_GetMEC` with the DIAG

Also the parameter `SOH_RunnableSOHMainPeriod` must be configured with the right periodicity for this runnable and the parameter `SOH_PermitExecution_Mask` with the appropriate byte to process the PermitExecution information outputed by SUSD.

3.2.4 Time Manager

SUM_SOH implements a periodic runnable (Runnable_SOH_Timing) that calls the OS API GetCounterValue() to read the counter Rte_TickCounter and pass it to:

- the Time Window Filter
- the ShutDown Delay Detector.

The periodicity of this runnable is configurable through the parameter SOHGeneral/SOH_RunnableSOHTimingPeriod.

Please note that this periodicity shall be less than 1 second in order to assure the proper increment of the time window timer and the shutdown delay timer.

3.2.5 Time Window Filter

SUM_SOH needs to update the time windows value when the relative filtering is started.

The calculation of the elapsed time is based on the ticks of a counter read periodically by the Time Manager Runnable (see 3.2.4).
The difference between the current ticks value and the previous one is converted in seconds: at least 1 second should elapse to increase a time window.
The previous value is initialized to 0 and reset to 0 when a new detection cycle starts.
If the previous value is 0, the difference is not computed.

3.2.6 Reliability Monitor

Reset Reliability Monitor

To integrate the Reset Reliability Monitor you must schedule the following Runnables in a Data Receive Event OS Task :

- `Runnable_SOHResetReliabilityMonitor_ImmeDelayedReset`
- `Runnable_SOHResetReliabilityMonitor_DiagReset`

Four NvM blocks must be created in the NvM :

DID_ImmeReset	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMImmeResetDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_DelayedReset	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMDelayedResetDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DiagResetCntr	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMDiagResetCntr
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Adaptable NVM

DID_DiagResetCntr	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMDiagResetDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

Interfaces that must be connected with the other SUMs :

- `SUSD_SOH_ResetOccured` with the SUSD relative Port
- `DIAG_ECUResetRequest` with the DIAG relative Port
- `NvMService` with the NvM relative Port

Memory Status Reliability Monitor

To integrate the Memory Status Reliability Monitor you must schedule the following Runnable in a Data Receive Event OS Task :

- `Runnable_SOH_ERRH_BswEvent`

And you shall use the following CallBack Function within OS ErrorHook(s):

- `SOH_OS_Protection_ErrorHook_CallBack`

Nine NvM blocks must be created in the NvM :

DID_NvMStat	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMStatDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_NvMAct	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMActDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_NvMPriv	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMPrivDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_ExcepOS	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_ExcepOSDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_ServOS	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_ServOSDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_StackOS	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_StackOSDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_MemProtOS	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_MemProtOSDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_TimeProtOS	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_TimeProtOSDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_ErrOS	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_ErrOSDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

To enable, disable the relative NvM error class you must configure the following configuration parameters:

Configuration Parameter	Detector
SOH_StatNvM_SupportEnabled	Status Failure/Integrity
SOH_ActNvM_SupportEnabled	Action Failure
SOH_PrivNvM_SupportEnabled	Privilege

To enable, disable the relative OS error class you must configure the following configuration parameters:

Configuration Parameter	Detector
SOH_ExcepOS_SupportEnabled	Hardware Protection
SOH_ServOS_SupportEnabled	Service Protection / Trusted Function
SOH_StackOS_SupportEnabled	Stack Fault
SOH_MemProtOS_SupportEnabled	Memory Protection
SOH_TimeProtOS_SupportEnabled	Time Protection
SOH_ErrOS_SupportEnabled	Other OS Errors

Interfaces that must be connected with the other SUMs :

- `ERRH_E_BswEvent` with the ERRH relative Port

Communication Reliability Monitor

Event-Triggered Runnables to schedule

To integrate the Communication Reliability Monitor you must schedule the following Runnables in a Data Receive Event OS Task:

- `Runnable_SOH_CommunicationReliabilityMonitor_ClearDiag()`

- `Runnable_SOH_CommunicationReliabilityMonitor_DiagSession()`
- `Runnable_SOH_CommunicationReliabilityMonitor_AcceptDiagSession()`
- `Runnable_SOH_CommunicationReliabilityMonitor_SecurityAccess()`
- `Runnable_SOH_CommunicationReliabilityMonitor_MacFailType123()`
- `Runnable_SOH_CommunicationReliabilityMonitor_SecP()`

Periodic Runnables to schedule

To integrate the Communication Reliability Monitor you must schedule the following Runnables in a Time Event OS Task:

- `Runnable_SOH_CommunicationReliabilityMonitor_MacFailType3Delay()`

NvM blocks to create

To integrate the Communication Reliability Monitor you must create the following NvM blocks in the NvM:

DID_ClearDiagInfo	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMClearDiagInfoDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_Reject_ClearDiag	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMRejectClearDiagDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_DiagSession	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMDiagSessionDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_Accept_DiagSession	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMAcceptDiagSessionDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_SecurityAccess	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMSecurityAccessDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_MACfail_TenAct	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMMACfailTenActDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_ReplayCount	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMReplayCountDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DID_SecP	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMSecPDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

DiagSessionCntr	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMDiagSessionCntr
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

Accept_DiagSessionCntr	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMAcceptDiagSessionCntr
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

SecurityAccessCntr	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMSecurityAccessCntr
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

Interfaces to connect

To integrate the Communication Reliability Monitor you must connect the following interfaces with the other SUMs:

- `SSFM_ClearDiagReq` with the SSFM relative Port
- `DIAG_SOH_DiagSessionRequest` with the DIAG relative Port
- `DiagnosticReprogrammingRequest` with the BswM relative Port
- `SSC_SOH_SecurityAccessError` with the SSC relative Port
- `SSC_E_SecureMessageMonitoringEvents` with the SSC relative Port
- `RP_ErrorStatus` with the BswM relative Port

Watchdog Manager Reliability Monitor

To integrate the Watchdog Manager Reliability Monitor you must schedule the following Runnables in an OS Task :

- `Runnable_WdgMReliabilityMonitor_FailedLocalSupervision_<Supervised_Entity_name>`
- `Runnable_SOH_BswEventsMonitor_NvM_DEM_Errors_Detector`
- `Runnable_SOH_WdgMReliabilityMonitor_ERRHFailedLocalSupervision`

Three NvM block must be created in the NvM :

NvMBlock_SOH_LocalSupervisionFailures_DID_counter	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_LocalSupervisionFailuresDID_counter
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

NvMBlock_SOH_WdgMDEMErr_DID_counter	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_WdgMDEMErr_DID_counter
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

NvMBlock_SOH_ERRHLocalSupervisionFailures_DID_counter	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_ERRHLocalSupervisionFailures_DID_counter
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

Interfaces that must be connected with:

- `SOH_WdgM_IndividualMode<1...n>` with the WdgM corresponding WdgM _LocalMode Port interface.
- `ERRH_E_BswEvent` with the ERRH corresponding Port
- `SOH_WdgM_IndividualMode_Errh` with the WdgM corresponding WdgM _LocalMode Port interface.
- `NvMService` with the NvM corresponding Ports

Watchdog Manager Configuration

Look section 3.2.8 for details.

Platform Reliability Monitor

NvM blocks to create

To integrate the Platform Reliability Monitor you must create the following NvM blocks in the NvM:

ShutDownCntr	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMShutDownCntr
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Adaptive NVM

DID_ShutDown	
NvMNvBlockLength	2
NvMRamBlockDataAddress	Rte_ArPim_SUM_SOH_NvMShutDownDID
NvMSelectBlockForReadAll	TRUE
NvMSelectBlockForWriteAll	TRUE
Configure as	Secure Cumulative NVM

Interfaces to connect

To integrate the Platform Reliability Monitor you must connect the following interfaces with the other SUMs:

- `PNC_SUSD_ShutdownPermission` relative port
- `ShutdownNotificationToSoh()` callback is configured as the first action in BswM shutdown action list.

Shutdown Delay Monitoring: ENTRY POINT

The RTA-SUM SOH Platform Reliability Monitor shall check the timing of the shutdown procedures. The monitor detects:

1. specific conditions that address the intention to get a shutdown (*REQUEST of shutdown*) and
2. specific notification that the appointed module has taken in charge the procedure (*EXECUTION of shutdown*)

The *REQUEST of shutdown* requires a correct communication among RTA-SUMs SUSD, PNC and SOH. RTA-SUM SOH triggers its Shutdown detector when it receives the shutdown request from RTA-SUM SUSD through `SUSD_PermitExecution` interface as follows:

- No Wake-up request with wake-up bit to FALSE(SOH functionalities are not required);
- Shutdown target bit to OFF

Besides No activity will have to be present, for any PN. RTA-SUM PNC provides this information to RTA-SUM SOH is the by interface `PNC_ShutdownPermission`

When The RTA-SUM SOH shutdown detector takes over these conditions, it starts all mechanisms (including timers) for starting the Shutdown Delay Monitoring.

Shutdown Delay Monitoring: EXIT POINT

The shutdown monitor detects the *EXECUTION of shutdown* when the callback `ShutdownNotificationToSoh()` is invoked by BswM ActionList for shutdown operations. A sample about how to configure a shutdown action list for BswM is shown as follows:

The action list order is not striclty mandatory as in figure 3.5, but it is strongly suggested. **What is mandatory** for configuring the BSW in the right way, is that: *The RteStop action must come after than action that notifies the RTA-SUM SOH*; Otherwise The callout will never be executed. The RTA-SUM SOH notification action should be configured in a similar way to the following one:

Once RTA-SUM SOH will receive this notification, it will disable the shutdown delay monitoring timer and it will take the current time from OS. The value comparison will determine if the shutdown procedure has taken more time than allowed.
The delay monitoring timer can be blocked also by a wake-up occurrence during the shutdown delay monitoring (while timer still has not expired): if interfaces `SUSD_PermitExecution` or `PNC_ShutdownPermission` do not match shutdown conditions anymore, the RTA-SUM SOH will disable AND reset the shutdown delay monitoring timer to the initial value, blocking the shutdown monitoring. In this case no notification is expected from BSW.

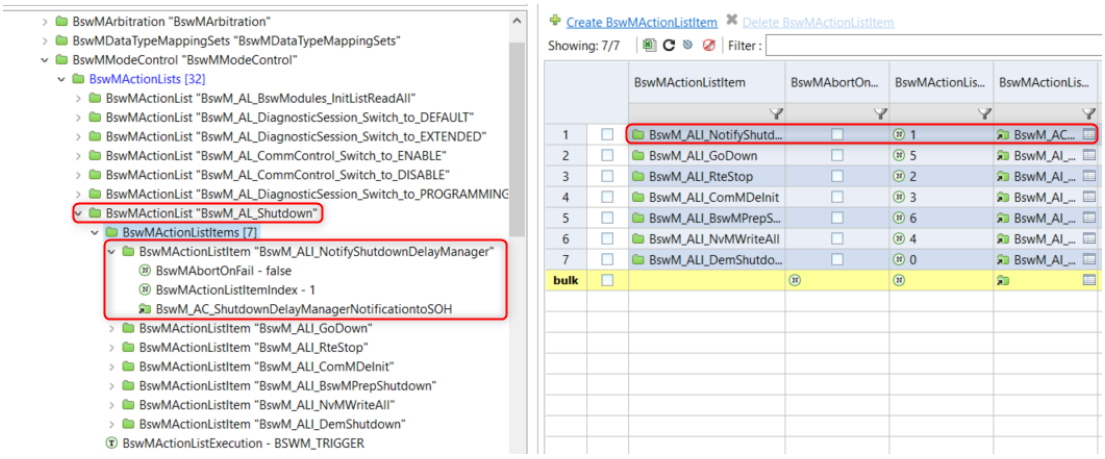


Figure 3.5: Shutdown BswM action list sample

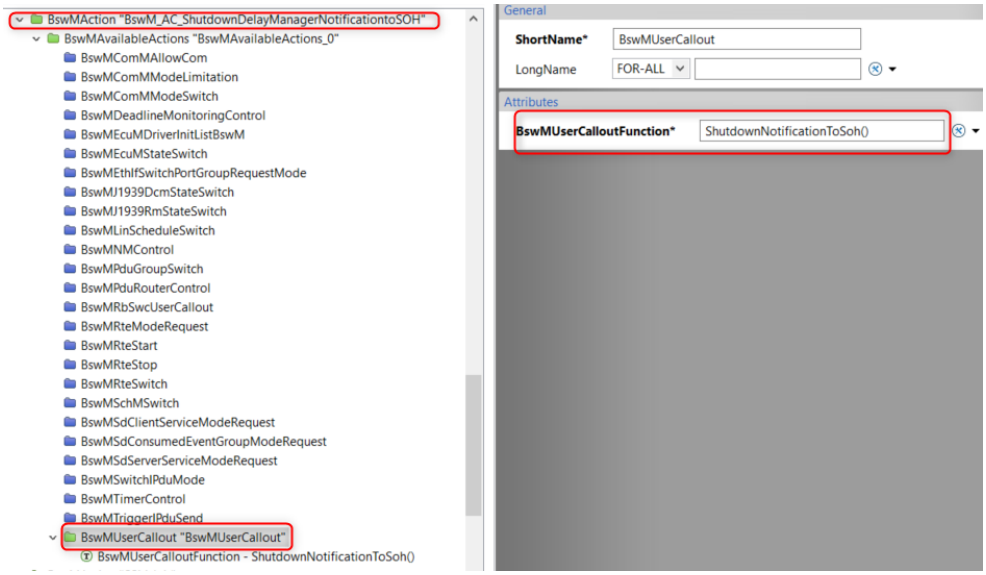


Figure 3.6: Shutdown BswM action list sample

NOTE: In case of

1. a continuous shutdown signals are being sent to RTA-SUM SOH
2. RTA-SUM SOH Platform Reliability Monitor has already detected the request and it is working for monitoring the progresses
3. Shutdown notification is never provided to RTA-SUM SOH (for any reason)

The RTA-SUM SOH will count just only one failure in the Platform Reliability Monitor for the Shutdown delay monitoring. This means that The RTA-SUM SOH will activate the Platform Reliability Monitor just when a change from normal execution request to a shutdown request is provided to it AND the shutdown notification OR wake-up request is provided to it during the shutdown delay monitoring.

3.2.7 DID Counters

The user must configure the DcmDspData and DcmDspDidSignals properly in the DCM for each detector. An example is depicted in Figure 3.7.

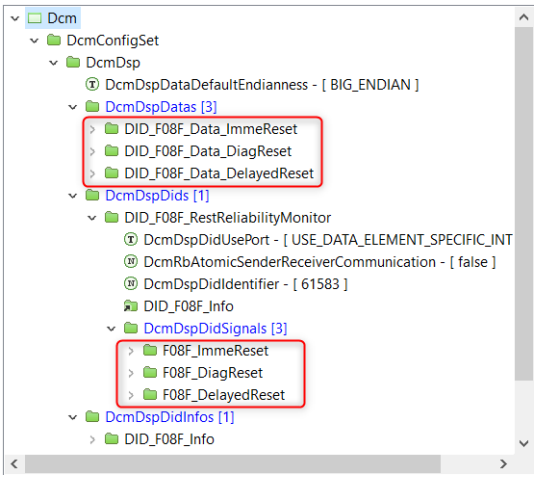


Figure 3.7: example of DID Configuration

The NvM blocks and Dcm reference configuration must to be linked to the correct ones in the BSW for each detector. An example is depicted in Figure 3.8.

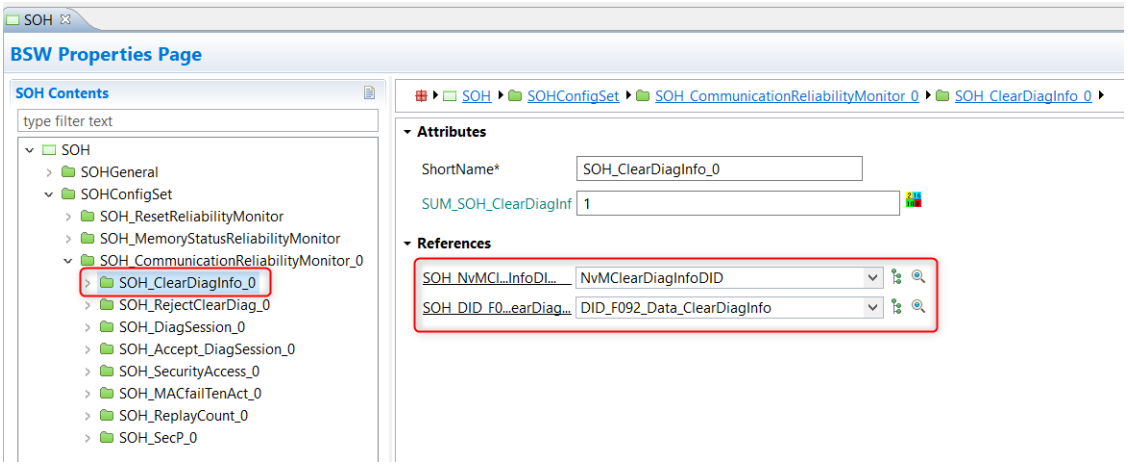


Figure 3.8: example of reference configuration to BSW

To update the DID Counters values in the DCM you must schedule the following Runnables in an operation invoked event OS Task :

- `Runnable_SOH_DID_F08F_Data_ImmeReset_Cntr`
- `Runnable_SOH_DID_F08F_Data_DiagReset_Cntr`
- `Runnable_SOH_DID_F08F_Data_DelayedReset_Cntr`

- Runnable_SOH_DID_F08F_CheckRead
- Runnable_SOH_DID_F090_Data_StatNVM_Cntr
- Runnable_SOH_DID_F090_Data_ActNVM_Cntr
- Runnable_SOH_DID_F090_Data_PrivNVM_Cntr
- Runnable_SOH_DID_F090_Data_ExcepOS_Cntr
- Runnable_SOH_DID_F090_Data_ServOS_Cntr
- Runnable_SOH_DID_F090_Data_StackOS_Cntr
- Runnable_SOH_DID_F090_Data_MemProtOS_Cntr
- Runnable_SOH_DID_F090_Data_TimeProtOS_Cntr
- Runnable_SOH_DID_F090_Data_ErrOS_Cntr
- Runnable_SOH_DID_F090_CheckRead
- Runnable_SOH_DID_F091_Data_ShutDown_Cntr
- Runnable_SOH_DID_F091_CheckRead
- Runnable_SOH_DID_F092_Data_ClearDiagInfo_Cntr
- Runnable_SOH_DID_F092_Data_Reject_ClearDiag_Cntr
- Runnable_SOH_DID_F092_Data_DiagSession_Cntr
- Runnable_SOH_DID_F092_Data_Accept_DiagSession_Cntr
- Runnable_SOH_DID_F092_Data_SecurityAccess_Cntr
- Runnable_SOH_DID_F092_Data_MACfail_TenAct_Cntr
- Runnable_SOH_DID_F092_Data_ReplayCount_Cntr
- Runnable_SOH_DID_F092_Data_SecP_Cntr
- Runnable_SOH_DID_F092_CheckRead
- Runnable_SOH_DID_F093_Data_LocalStatFail_Cntr
- Runnable_SOH_DID_F093_Data_Data_WdgM_DEMErr_Cntr
- Runnable_SOH_DID_F093_Data_ERRHSupFail_Cntr
- Runnable_SOH_DID_F093_Data_ERRH_Local_Supervision_Status
- Runnable_SOH_DID_F093_CheckRead

SUM_SOH will populate the DIDs value in the BIG_ENDIAN format. This is the expected endianness of the UDS layer. Make sure to set DcmDspDataUsePort to CLIENT_SERVER (means that the application, SUM_SOH, is its manager).

3.2.8 WdgM

The SUM_SOH releases for SDV 2.0 works as a Supervidsed Entity in order to be monitored by WdgM with an **Alive Supervision** method. This is possible thanks to Client-Server port **RP_SOH_WdgM_AliveSupervision**, with port interface `WdgM_LocalSupervision`. When Checkpoint is reached, SUM_SOH sends a notification (the Alive Indication) to WdgM each time that `Runnable_SOH_Main` is executed (Periodic Runnable).

The Alive Supervision requires the configuration of *Supervsion Cycles* as time slots where the Alive Indication can be accepted for updating the Alive counting.

The duration of each Supervision Cycle represents the schedule time for `WdgM_MainFunction` and it is configured thanks to parameters `WdgMSupervisionCycle` in WdgMConfigSet/WdgMMode.
Supervision Cycle length can be configured in order to occurr more frequently than Alive Indication or less, acording to own speicifc needs and AR specification about WdgM.

The configuration parameters to configure an Alive Supervision are represented in figure 3.9

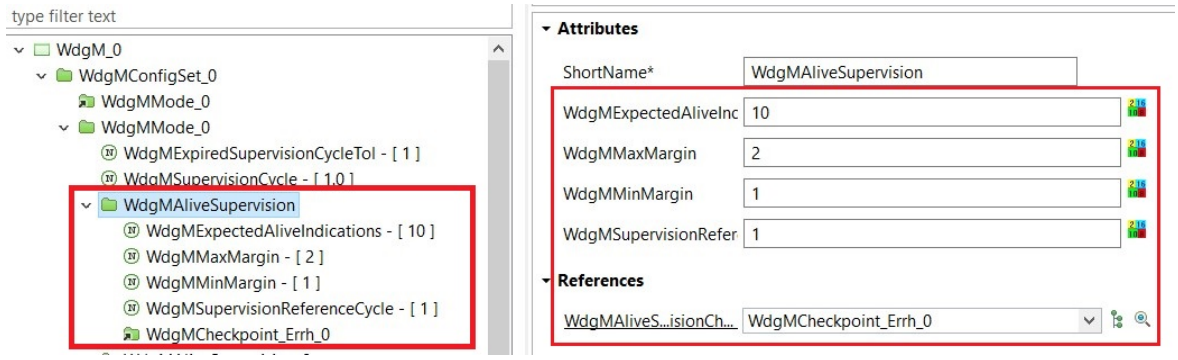


Figure 3.9: Configuration Parameters for Alive Supervision

The configuration hints are the following:

- `WdgMExpectedAliveIndications`: This parameters shall contain the expected alive indication that can be derived from `WdgMSupervisionCycle` divided by `Runnable_ERRH_MainFunction` periodicity in addition to `WdgMSupervisionReferenceCycle`;
- `WdgMMaxMargin`: This parameters defines the number of extra Alive Indication that are tollerated;
- `WdgMMinMargin`: This parameters defines the number of less Alive Indication that are tollerated;
- `WdgMSupervisionReferenceCycle`: This parameter is used to define the number of Supervision Cycles where the Alive Indications can arrive;
- `WdgMAliveSupervisionCheckpointRef`: This is a reference to the Checkpoint that shall be reached to produce Alive Indication;

3.2.9 RID 0x02C0 Clear SOH Logs

To clear the SOH metrics the `Runnable_SOH_RIDClearLogs` is synchronicity invoked through an Operation Invoked Event in the DCM execution context.

It implements the Server part of the ClientServer Interface `RoutineServices_Clear_SUM_SOH_logs`.

To obtain that interface you must configure, in the DCM container, a DcmDspRoutine named "Clear_SUM_SOH_logs" that implements the RID 0x02C0 specifications described in the GB6001.

If at least one metric could not be written in the NvM due to an ongoing write activity or to an error during the NvM writing, the Routine will return a `DCM_E_GENERALREJECT` error code.

When the DCM calls the RID with an invalid Option Record values or an Option Record value related to a non-enabled detector/monitor, the Routine will return a `DCM_E_REQUESTOUTOFRANGE` error code.

The **Routine Status Record** is always set to the value **0x4** (Autosar supported, SOH integrated and RID executed). If the SUM_SOH is not integrated in the system, the implementation of this routine and its status record value is in charge of the customer.

When the DCM calls the RID with Option Record values 0xF094 or 0xFFFF, the SUM_SOH invokes a client operation service in the SUM_ERRH (Acting as a server) to clean all the metrics related to SUM_SOH Local Supervision Detector implemented at the ERRH side. To complete the RID 0x02C0 routine the SUM_SOH shall interact with the SUM_ERRH through the Client-Server interface SOH_ERRH _RID_Clear_Logs where SUM_ERRH acts as a Server.

Please connect the SUM_SOH R-Port **RP_SOH_ERRH_ClearRID0x02C0** with the relative one's of the SUM_ERRH.

3.2.10 RTE RteSwComponentType configuration

The component SUM_SOH is delivered as source and not as object code. To configure in the right way the SUM_SOH in your project, you need to create an **RteSwComponentType** that references the SUM_SOH component and its SwcImplementation. Please check the Figure 3.10

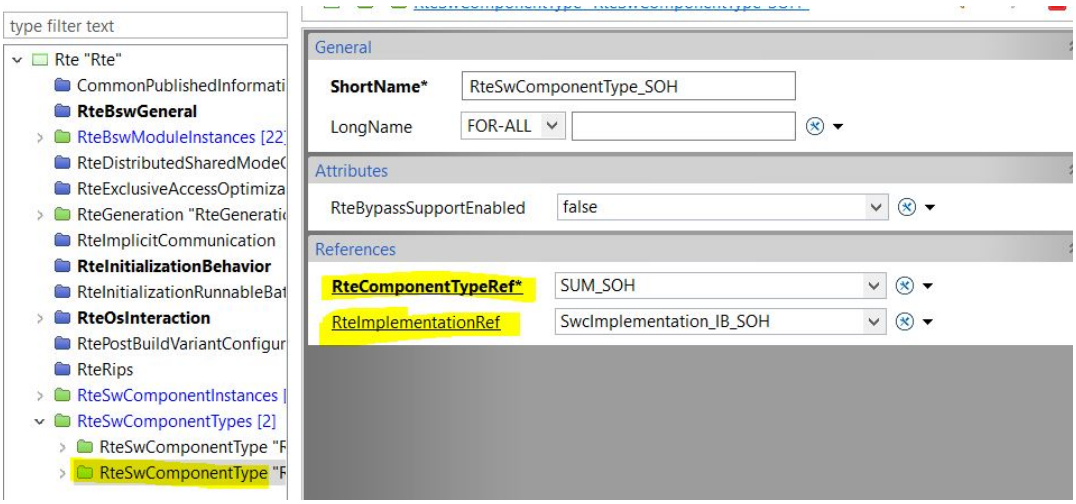
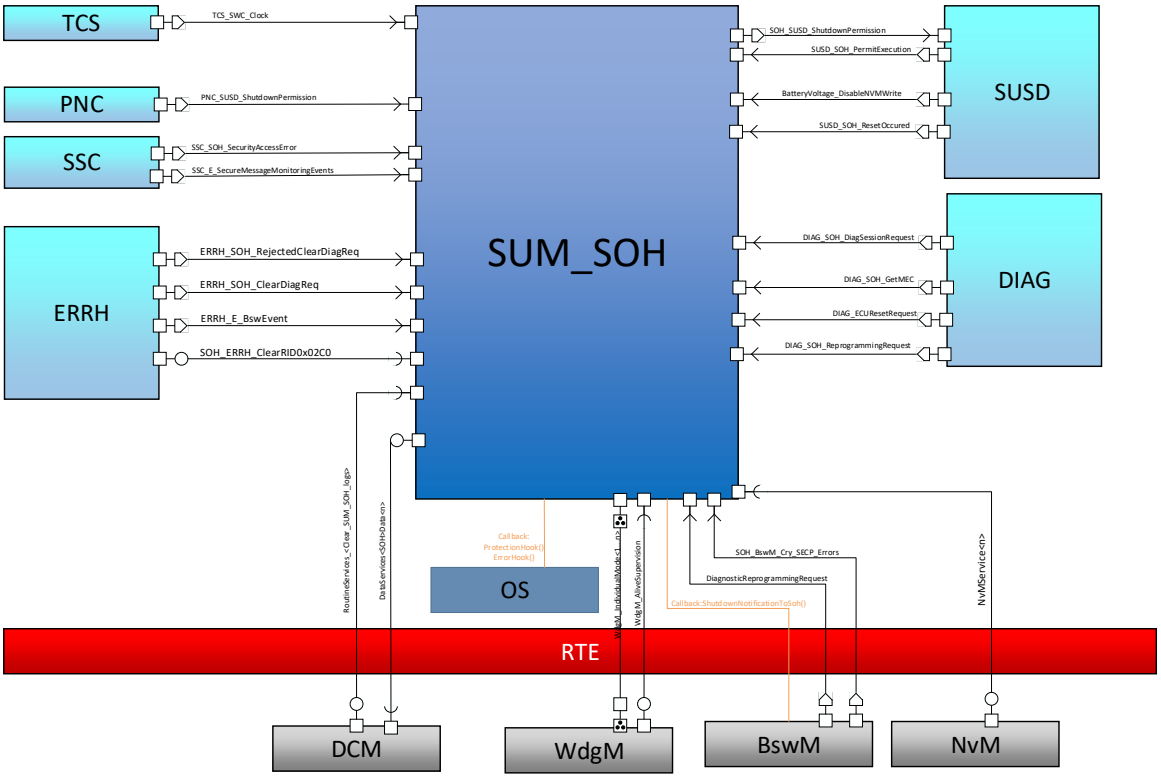


Figure 3.10: Configuration of RteSwComponentType

4 Generic Component Interface



Port Interface	Port Type	P-Port	R-Port	Description
ERRH_E_BswEvent	S / R	SUM_ERRH	SUM_SOH	Reception of DEM events which are routed from SUM_ERRH to the SUM_SOH for processing.

Port Interface	Port Type	P-Port	R-Port	Description
SUSD_SOH_PermitExecution	S / R	SUM_SUSD	SUM_SOH	Permit Execution interface to indicate that operating conditions are correct for the SUM_SOH to execute its functions.
PNC_SUSD_ShutdownPermission	S / R	SUM_PNC	SUM_SOH	Notification to the SUM_SOH of any active or not active partial networks.
SSC_E_ SecureMessageMonitoringEvents	S / R	SUM_SSC	SUM_SOH	Notification of MAC failures along with their types.
DIAG_ECUResetRequest	S / R	SUM_DIAG	SUM_SOH	Notification of a service 0x11 tester-initiated request for ECU Reset (not programming).
SUSD_SOH_ResetOccured	S / R	SUM_SUSD	SUM_SOH	Notification to the SUM_SOH that a reset occurred. The type of reset is also communicated to the SUM_SOH. Type of resets may be delayed or immediate.
SSFM_ClearDiagReq	S / R	SUM_SSFM	SUM_SOH	Notification to the SUM_SOH that a service 0x04 or a service 0x14 request was accepted or rejected.

Port Interface	Port Type	P-Port	R-Port	Description
DIAG_SOH_DiagSessionRequest	S / R	SUM_DIAG	SUM_SOH	Notification to the SUM_SOH that a reprogramming request took place (service 0x10 subfunction 0x02).
SSC_SOH_SecurityAccessError	S / R	SUM_SSC	SUM_SOH	Notification to the SUM_SOH of the occurrences of service 0x27 failures.
BatteryVoltage_DisableNVMWrite	S / R	SUM_SUSD	SUM_SOH	Notification to the SUM_SOH from SUM_SUSD that vehicle battery voltage is too low to perform NVM writes.
DIAG_GetMEC	S / R	SUM_DIAG	SUM_SOH	This interface is used by the SUM_DIAG to communicate to SUM_SOH to value of the MEC.
DataServices<SOH>Data<n>	C / S	DCM	SUM_SOH	To support the collection of Diagnostic DIDs that are part of the SUM_SOH functionalities.
DiagnosticReprogrammingRequest	Mode Switch	BswM	SUM_SOH	Notification of a request to shutdown to reset as part of programming to jump to boot.

Port Interface	Port Type	P-Port	R-Port	Description
SOH_WdgM_IndividualMode<1...n>	Mode Switch	WdgM	SUM_SOH	Notification of WdgM local supervision failures occurrence for all supervised entities excluding the SUM_SOH and the SUM_ERRH. These interfaces replace the SOH_BswM_WdgM_Individ-ualMode<1...n> interfaces described in GB8002.
SOH_WdgM_IndividualMode_Errh	Mode Switch	WdgM	SUM_SOH	Notification of the WdgM SUM_ERRH local supervision failures occurrences. This interface replaces the SOH_BsWM_WdgM_ERRHMode interface described in GB8002.
RP_ErrorStatus	Mode Switch	BswM	SOH	This interface is used by the BswM to notify SUM_SOH of SecP errors occurrences.
NvMService	C / S	NvM	SOH	The interface provides services for the management and maintenance of NV data related to SOH Metrics.
TCS_SWC_Clock	C / S	TCS	SOH	Provides Global Time for timestamp (unit 100ms).

Port Interface	Port Type	P-Port	R-Port	Description
RoutineServices_Clear_SUM_SOH_logs	C / S	DCM	SOH	The interface provides services for the clear of the SOH Metrics.
WdgM_LocalSupervision	C / S	WdgM	SUM_SOH	Notify the checkpoint reachement for implementing the Local Supervision.
SOH_ERRH_ClearRID0x02C0	C / S	SUM_ERRH	SUM_SOH	The interface provides service for starting clear, all the metrics related to SUM_SOH Local Supervision Detector implemented at ERRH side.
SOH_SUSD_ShutdownPermission	S / R	SUM_SOH	SUM_SUSD	Shutdown Permission interface to indicate that operating conditions are correct for the SUM_SOH to execute the shutdown.

5 Contact, Support and Problem Reporting

5.1 Technical Support

Technical support is available to all RTA users with a valid support contract. If you do not have a valid support contract, please contact your regional sales office (see below).

The best way to get technical support is by email. Any problems or questions about the use of the product should be sent to:

- rta.hotline@etas.com

In either case, it is helpful if you can provide any of the following relevant information when contacting technical support:

- Your support contract number
- The version(s) of the ETAS tool(s) you are using
- The error messages or return codes you received (if any)
- The operation that resulted in the error

5.2 ETAS Global Headquarters

ETAS GmbH Borsigstrasse 24	Phone:	+49 711 3423-0
70469 Stuttgart	Fax:	+49 711 3423-2106
Germany	www:	www.etas.com

5.3 ETAS Local Sales and Support Offices

Contact details for your local sales office and local technical support team (where available) can be found on the ETAS web site:

ETAS subsidiaries:	https://www.etas.com/en/contact.php
ETAS technical support:	https://www.etas.com/en/hotlines.php